



Matrices · Worksheet 1

Core Pure 1 Chapter 6: Matrices. Answer all questions. Show your working clearly. Total: 83 marks. Time: 105 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are four sections, one for each topic in Chapter 6. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- **I** denotes the identity matrix and **0** the zero matrix. Give exact answers, using fractions where necessary. Where a question asks you to find a determinant or an inverse, show the method: answers written down from a calculator will not gain full marks.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Matrix arithmetic and multiplication	19	
B Determinants and inverses of 2×2 matrices	20	
C Determinants and inverses of 3×3 matrices	19	
D Solving systems of linear equations	20	
Roar-level challenge	5	
Total	83	

Section A Matrix arithmetic and multiplication

19 marks

1. Given that $\mathbf{A} = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 1 & 5 \\ -2 & 0 \end{pmatrix}$, find $2\mathbf{A} - 3\mathbf{B}$. [2]

.....

.....

2. For the matrices **A** and **B** in Question 1, find **AB** and **BA**, and comment on your answers. [3]

.....

.....

.....

.....



3. $\mathbf{C} = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix}$ and $\mathbf{D} = \begin{pmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 4 \end{pmatrix}$. [3]

State the orders of $\mathbf{CD}$ and $\mathbf{DC}$, and find $\mathbf{CD}$.

.....

.....

.....

.....

4. $\mathbf{A} = \begin{pmatrix} a & 2 \\ -1 & 3 \end{pmatrix}$, where a is a constant. [4]

(a) Given that $\mathbf{A}^2 = \begin{pmatrix} -1 & 8 \\ -4 & 7 \end{pmatrix}$, find the value of a .

(b) Hence show that $\mathbf{A}^2 - 4\mathbf{A} + 5\mathbf{I} = \mathbf{0}$.

.....

.....

.....

.....

.....



5. $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$.

(a) Find $\mathbf{A}^2$ and $\mathbf{A}^3$.

[2]

.....

.....

.....

(b) Show that $\mathbf{A}^2 = 5\mathbf{A} - 6\mathbf{I}$.

[2]

.....

.....

(c) Hence express $\mathbf{A}^4$ in the form $p\mathbf{A} + q\mathbf{I}$, where p and q are integers to be found.

[3]

.....

.....

.....

.....

(Total for Question 5 is 7 marks)



Section B Determinants and inverses of 2×2 matrices

20 marks

6. Find the determinant of $\begin{pmatrix} 5 & 3 \\ 4 & 2 \end{pmatrix}$, and state whether the matrix is singular or non-singular. [2]

.....

.....

7. Find the inverse of the matrix $\begin{pmatrix} 3 & 4 \\ 2 & 5 \end{pmatrix}$. [3]

.....

.....

.....

8. Find the values of k for which the matrix $\begin{pmatrix} k & 3 \\ 4 & k+1 \end{pmatrix}$ is singular. [3]

.....

.....

.....

9. $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 5 & 3 \end{pmatrix}$ and $\mathbf{AB} = \begin{pmatrix} 1 & 2 \\ 3 & 7 \end{pmatrix}$. [4]
Find $\mathbf{A}^{-1}$, and hence find the matrix $\mathbf{B}$.

.....

.....

.....

.....

.....



10. $\mathbf{M} = \begin{pmatrix} k & 7 \\ -1 & k-4 \end{pmatrix}$, where k is a real constant.

(a) Show that $\mathbf{M}$ is non-singular for all values of k . [3]

.....

.....

.....

(b) Find $\mathbf{M}^{-1}$ in terms of k . [2]

.....

.....

.....

(c) Given that $k = 2$, find the vector $\mathbf{v}$ such that $\mathbf{M}\mathbf{v} = \begin{pmatrix} 16 \\ -5 \end{pmatrix}$. [3]

.....

.....

.....

(Total for Question 10 is 8 marks)

**Section C Determinants and inverses of 3×3 matrices**

19 marks

11. Find the determinant of the matrix $\begin{pmatrix} 2 & 0 & 1 \\ 3 & 1 & -1 \\ 0 & 4 & 2 \end{pmatrix}$. [2]

.....

.....

.....

12. Find the values of k for which the matrix $\begin{pmatrix} 1 & 2 & k \\ 0 & k & 1 \\ 3 & 1 & 2 \end{pmatrix}$ is singular. [3]

.....

.....

.....

.....

13. $\mathbf{A} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 0 & 1 \end{pmatrix}$. [3]

Find the matrix of cofactors of $\mathbf{A}$, and use it to find $\det \mathbf{A}$.

.....

.....

.....

.....



14. Find the inverse of the matrix $\mathbf{B} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}$. [4]

.....

.....

.....

.....

.....

.....

15. $\mathbf{M} = \begin{pmatrix} 1 & a & 0 \\ 0 & 1 & a \\ a & 0 & 1 \end{pmatrix}$, where a is a real constant.

- (a) Show that $\det \mathbf{M} = 1 + a^3$. [2]

.....

.....

.....

- (b) State the value of a for which $\mathbf{M}$ is singular. [1]

.....

- (c) Given that $\mathbf{M}$ is non-singular, find $\mathbf{M}^{-1}$ in terms of a . [4]

.....

.....

.....

.....

.....

.....

(Total for Question 15 is 7 marks)



Section D Solving systems of linear equations

20 marks

16. Use a matrix method to solve the simultaneous equations

[2]

$$3x + 4y = 10, \quad 2x + 5y = 9.$$

.....

.....

.....

$$17. \mathbf{N} = \begin{pmatrix} 1 & -1 & 2 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \text{ and } \mathbf{N}^{-1} = \frac{1}{7} \begin{pmatrix} 1 & 3 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 3 \end{pmatrix}.$$

[3]

Use $\mathbf{N}^{-1}$ to solve the equations

$$x - y + 2z = 9, \quad 2x + y = 0, \quad y + z = 1.$$

.....

.....

.....

.....

18. Find the value of k for which the equations

[3]

$$x + 2y + z = 1, \quad 2x + ky + 3z = 4, \quad x + y + 2z = 3$$

do **not** have a unique solution.

.....

.....

.....

.....



19. Three planes have equations

[4]

$$x + y + z = 2, \quad x - y + 2z = 1, \quad 3x + y + 4z = 6.$$

- (a) Show that the planes do not meet at a unique point.
- (b) Show that the equations are inconsistent, and describe the geometrical arrangement of the three planes.

.....

.....

.....

.....

.....

.....



20. Consider the system of equations

$$\begin{aligned}x + y + z &= 6 \\2x - y + kz &= b \\x + 2y + 3z &= 14\end{aligned}$$

where k and b are constants.

(a) Find, in terms of k , the determinant of the matrix of coefficients. [2]

.....

.....

.....

(b) Given that $k = 1$ and $b = 3$, solve the equations. [3]

.....

.....

.....

.....

(c) Given that $k = -4$ and $b = 3$, show that the equations are inconsistent, and interpret this geometrically. [3]

.....

.....

.....

.....

(Total for Question 20 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

21. $\mathbf{A} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$

[5]

Show that $\mathbf{A}^2 = \mathbf{A} + 2\mathbf{I}$. Hence, without finding any determinants or cofactors, find $\mathbf{A}^{-1}$.

.....

.....

.....

.....

.....

.....

.....

.....

END OF WORKSHEET

Total: 83 marks